CFD v2 — OpenFOAM Air-Side Heater Unit Cell: Full Technical Report

Project: Combined Microreactor — Phase 1 (physics / derating), Stream A **Reference design:** Westinghouse eVinci-class heat-pipe microreactor, 15 MWth / 5 MWe, open-air Brayton PCS **Date:** 2026-07-08 **Solver:** OpenFOAM (ESI) v2406, `rhoSimpleFoam`, steady compressible RANS + k-ω SST **Status:** complete and validated; merged to `main`

> **Purpose of this document.** This is the self-contained reproduction and reference record for > the OpenFOAM campaign that produced the **v2 derating curve**. It states every specification, > parameter, dictionary value, geometry definition, boundary condition, property model, numerical > setting, and result used in the run, plus the methodology and honest limitations. A subsequent > agent should be able to (a) rebuild the case from scratch, (b) re-run or extend it, or (c) reuse > the numbers as a validated reference — without re-deriving anything. Where a value is a *measured* > CFD output it is flagged as such; where it is an *input* or a *modelling choice* it is flagged too.

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0. Executive summary

The open-air Brayton power conversion system of the reference microreactor rejects/absorbs reactor heat through an **air-cooled finned-tube heater**. The v1.5 cycle model had to *assume* how that heater's conductance `UA` scales with air mass flow ($UA \propto \dot{m}^{0.6}$). This campaign replaced that assumption with a physics simulation:

- Built an OpenFOAM CFD model of one staggered finned-tube **unit cell** of the heater.
- Ran it at the cycle **design Reynolds number (Re = 8368)** and measured the air-side heat transfer.
- **Validated** the measured heat transfer against the published **Briggs–Young (1963)** correlation:

`Nu_CFD(LMTD) = 45.9` vs `Nu_BY = 54.1` → **−15.2 %, inside the ±20 % engineering gate → VALIDATED.**

- Adopted the now-validated Briggs–Young `Nu(Re)` law, converted it to a heater `UA(ṁ)` law

(modulated by fin efficiency), and injected it back into the v1.5 cycle model to regenerate the derating curve — the **v2 curve**.

Headline result: v2 penalty at 55 °C = **−8.7 %** vs v1.5's **−8.8 %** (a **+0.08-point delta**). **v2 confirms v1.5.** The derating is *robust* to the exact air-side heat-transfer law because the open-air Brayton's ambient-driven air mass flow varies only ~4–5 % over 25→55 °C, making the UA-scaling exponent a second-order effect. The desert penalty is dominated by compressor-inlet thermodynamics, not by the heater's heat-transfer law.

Durable asset produced: a **validated OpenFOAM air-side setup** that can be pointed at problems no textbook correlation covers (the intended next study: intake/exhaust recirculation).

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1. Physical problem and modelling scope

1.1 What is being modelled

The heater is a bank of horizontal, transversely-finned tubes carrying reactor heat (via sodium heat pipes / the primary loop) into the compressed air stream that then expands through the turbine. The CFD models **one representative tube** of that staggered bank as a periodic **unit cell** in crossflow, at the on-design air state (≈2 bar, 740 K inlet), with an **isothermal wall at 1033 K** standing in for the heat-pipe-fed tube surface. The quantity of interest is the **air-side convective heat-transfer coefficient** `h` (and hence `Nu`), from which the heater conductance `UA = h·A·η_o` and its mass-flow scaling are derived.

1.2 What is deliberately *not* modelled

- **No conjugate heat transfer / no solid conduction inside the tube or fin metal.** The wall is a

fixed-temperature boundary. Fin conduction is handled analytically afterward via a fin-efficiency factor (§5.3), not inside the CFD.

- **No radiation.**
- **No heat-pipe / primary-side physics.** The 1033 K wall is the interface to that side.
- **Single tube only** — the staggered-neighbour interleaving is approximated by periodic/symmetry

boundaries (see §4 and the fin-clipping limitation §7.1).

- **Steady state** (`rhoSimpleFoam`, `ddt = steadyState`). No transient / unsteady wake shedding.

1.3 Design operating point (the single validated point — "Option 1")

Quantity	Value	Origin
Air inlet static temperature `T_in`	**740 K**	cycle compressor-exit / heater-inlet air temperature
Isothermal tube wall temperature `T_wall`	**1033 K**	heat-pipe-fed tube surface (input BC)
Heater-side static pressure `p`	**2.0×10^5 Pa** (~2 bar)	post-compressor pressure ratio × ambient
Inlet approach velocity `U_in`	**6.3623 m/s** (+y)	set to hit the design Re
Reynolds number Re (tube-OD, min-flow-section mass flux)	**8368**	cycle design mass flux
Prandtl number `Pr`	**0.69** (explicit modelling choice)	high-T air

Only one CFD point was run ("Option 1"). Once this point validated heat transfer against the correlation, a multi-point Reynolds sweep would only have refined the `Nu(Re)` *shape* — which the sensitivity analysis (§6.3) shows is second-order to the derating result. See §8 (DECISIONS_LOG D12) for the full justification.

2. Geometry — staggered finned-tube unit cell (written out in full)

This is the exact geometry meshed in OpenFOAM. It is generated parametrically in Python (`geometry/finned_tube.py` for the dimensions and derived quantities; `geometry/make_stl.py` for the triangulated surface `tube.stl` that snappyHexMesh carves out of the background box).

2.1 Reference dimensions (`FinnedTube.REFERENCE`)

All lengths in metres unless noted. A single annularly-finned circular tube:

Symbol	Parameter	Value	mm
`d_o`	tube outer diameter	0.0254	**25.400 mm**
`fin_h`	fin height (radial, root→tip)	0.012	**12.000 mm**
`fin_t`	fin thickness (axial)	0.0005	**0.500 mm**
`fin_pitch`	fin centre-to-centre spacing (axial)	0.004	**4.000 mm**
`S_T`	transverse tube pitch = 2.00·d_o	0.0508	**50.800 mm**
`S_L`	longitudinal tube pitch = 1.75·d_o	0.044450	**44.450 mm**
`k_fin`	fin thermal conductivity (high-T alloy)	25.0 W/m·K	—

Derived radii: tube outer radius `r_t = d_o/2 = 12.700 mm`; **fin outer radius** `r_f = d_o/2 + fin_h = 24.700 mm`. Note `r_f (24.700 mm) > S_L/2 (22.225 mm)` — the fins are taller than the half-longitudinal-pitch. This is intrinsic to the reference design (fin height ≈ tube radius) and is the root of the fin-clipping limitation (§7.1).

2.2 Derived geometric quantities (used by the correlations and the UA law)

Computed from the closed forms in `geometry/finned_tube.py`, evaluated for `REFERENCE`:

Quantity	Formula (per fin pitch unless noted)	Value
Minimum free-flow area / pitch	`(S_T − d_o)·fin_pitch − 2·fin_h·fin_t`	8.9600×10^{-5} m²
Minimum free-flow area (3 pitches = domain A_min)	3 × above	2.6880×10^{-4} m²
Total air-side area / pitch	`2 fin faces + fin edge + exposed bare tube`	3.176778×10^{-3} m²
Total air-side area (3 pitches, analytical A_air)	3 × above	9.530335×10^{-3} m²

Hydraulic diameter	`4·A_min_perpitch·S_L / A_air_perpitch`	5.0148 mm
Fin efficiency at design h (≈130.3 W/m²K)	`tanh(mL_c)/(mL_c)`, Harper `L_c = fin_h + fin_t/2`, `m = √(2h/(k_fin·fin_t))`	**0.5334**

The closed forms (for reproduction): ` min_flow_area_per_pitch = (S_T − d_o)·fin_pitch − 2·fin_h·fin_t air_area_per_pitch = 2π(r_f² − r_t²) [two fin faces] + 2π·r_f·fin_t [fin outer edge] + π·d_o·(fin_pitch − fin_t) [exposed bare tube between fins] hydraulic_diameter = 4·min_flow_area_per_pitch·S_L / air_area_per_pitch fin_efficiency(h) = tanh(m·L_c)/(m·L_c), m = √(2h/(k_fin·fin_t)), L_c = fin_h + fin_t/2 `

2.3 The meshed unit cell (STL construction — `make_stl.py`)

The STL is a **single tube spanning 3 fin pitches** (`n_fins = 3`, `total_z = 3 × 4 mm = 12 mm`), built as a watertight triangulated surface with `n_theta = 64` circumferential facets:

- **Bare-tube cylinder** at radius `r_t`, emitted **only over the axial sub-intervals* between* fin

bands** (the tube surface inside a fin band is interior to the fin solid and must not be emitted, or snappyHexMesh sees a coincident interior patch).

- **3 annular fins**, each a thin disc of thickness `fin_t` centred at `z = (k+0.5)·fin_pitch` for

`k = 0,1,2` (i.e. z = 2, 6, 10 mm). Each fin = two flat annuli (`r_t → r_f`) at `z_c ± fin_t/2` (bottom face wound to give a −z outward normal, top face +z), plus a short cylinder at `r_f` forming the fin outer edge.

Axis convention (critical — resolves an earlier plan typo):

- **z = tube axis = spanwise.** Fins stack along z at `fin_pitch`. z is a symmetry/closure direction,

not streamwise.

- **y = streamwise.** Crossflow enters in +y.
- **x = transverse** (the S_T direction).

STL bounding box: `(2·r_f, 2·r_f, total_z) = (49.4 mm, 49.4 mm, 12 mm)`. The measured (fin-clipped) wetted area of the meshed `tube` patch is **9.0165 × 10^{-3} m²** (329 016 faces), i.e. 0.946× the analytical unclipped `A_air = 9.530 × 10⁻³ m²` — the deficit is the fin metal clipped by the streamwise box boundary (§7.1).

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3. Computational domain and mesh

3.1 Background block (`system/blockMeshDict`)

A single hex block sized to one unit cell, centred on the tube axis at (x,y) = (0,0):

Direction	Extent	Physical meaning	Length
x	−0.0254 … +0.0254	± S_T/2 (transverse)	50.8 mm
y	−0.022225 … +0.022225	± S_L/2 (streamwise)	44.45 mm

z	0.0 … 0.012	3 × fin_pitch (spanwise, tube axis)	12.0 mm

Base grid: `(40 × 34 × 9)` cells, `simpleGrading (1 1 1)` → base cell ≈ `d_o/20 ≈ 1.27 mm`.

Boundary patches (background):

Patch	Type	Face	Role
`inlet`	patch	y = ymin	air enters +y
`outlet`	patch	y = ymax	air exits
`cyclicX_neg` / `cyclicX_pos`	cyclic pair	x = xmin / xmax	transverse periodicity (tube bank in x)
`symZ_neg` / `symZ_pos`	symmetryPlane	z = zmin / zmax	spanwise closure at exact mid-fin-gap planes
`tube`	wall (from snappy)	STL surface	the finned-tube surface

Why symmetryPlane (not cyclic) in z: z = 0 and z = 12 mm sit at exact symmetry planes of the fin array (mid-gap between fins). The tube axis lies *along* z, so the wall crosses these planes; symmetry is layer-meshable there where cyclic is not, and is physically identical for this configuration. z is a **closure** direction, not a streamwise-cyclic bundle direction.

3.2 snappyHexMesh (`system/snappyHexMeshDict`)

Castellate + snap + add layers, carving the STL tube out of the background box.

- **Refinement:** surface `level (2 3)` → 0.16–0.32 mm on the general tube surface. Feature edges (`tube.eMesh`) at `level 3`. **Gap refinement** `gapLevel (4 1 4)` drives the thin 0.5 mm fins (tip + faces) to level 4 (~0.08 mm) so ≥4 cells span the fin thickness — without this the fin tips sit on 0.32 mm cells and hit y^+ ~7 at the shoulders.
- `nCellsBetweenLevels 2`, `resolveFeatureAngle 30`, `maxGlobalCells 4 000 000`.
- `locationInMesh (0.023 0.020 0.0005)` — a domain corner firmly in the air, outside the finned envelope (`|(0.023, 0.020)| = 0.0305 m > r_f = 0.0247 m`).
- **Boundary layers on `tube`:** `nSurfaceLayers 8`, `expansionRatio 1.2`, `relativeSizes true`, `finalLayerThickness 0.3` (thinner first cell ≈ 24 µm for y^+ margin), `minThickness 0.008`, `featureAngle 170` (wraps layers around convex fin edges/tips).

3.3 Feature extraction (`system/surfaceFeatureExtractDict`)

`extractFromSurface`, `includedAngle 150` (edges sharper than 30° kept). `nonManifoldEdges no`, `openEdges no` (tube ends lie on symmetry planes — no feature there).

3.4 Mesh quality limits (`system/meshQualityDict`)

Standard OpenFOAM defaults tuned for the wall-resolved layers: `maxNonOrtho 65` (relaxed 75), `maxBoundarySkewness 20`, `maxInternalSkewness 4`, `maxConcave 80`, `minVol 1e-14` (allows thin wall-resolved layer cells), `minTetQuality 1e-15`, `minTwist 0.02`, `minDeterminant 0.001`, `minFaceWeight 0.02`, `minVolRatio 0.01`, `nSmoothScale 4`, `errorReduction 0.75`.

3.5 Resulting mesh (measured — `log.checkMesh`, `constant/polyMesh/boundary`)

Metric	Value
Total cells	**3 288 913** (≈3.29 M)
Hexahedra / prisms / polyhedra	3 092 437 / 4 872 / 191 308
Max aspect ratio	22.44 — OK
Max non-orthogonality	**64.84** (avg 7.08) — OK
Max skewness	**3.05** — OK
checkMesh verdict	**Mesh OK**
`tube` patch faces (wetted area)	329 016 faces (9.0165×10^{-3} m²)
`inlet` / `outlet` faces	9 828 each

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4. Boundary conditions and physics setup

4.1 Field boundary conditions (`0.orig/`)

Air enters in +y at 740 K, ~2 bar; the finned tube is an isothermal 1033 K no-slip wall.

Field	internalField	`inlet`	`outlet`	`tube` **(wall)**	`cyclicX_neg/pos`	`symZ_neg/pos`
U [m/s]	(0 6.3623 0)	fixedValue (0 6.3623 0)	pressureInletOutletVelocity	noSlip	cyclic	symmetryPlane
T [K]	740	fixedValue 740	inletOutlet (740)	**fixedValue 1033**	cyclic	symmetryPlane
p [Pa]	2.0e5	zeroGradient	fixedValue 2.0e5	zeroGradient	cyclic	symmetryPlane
k [m²/s²]	0.1518	fixedValue 0.1518	inletOutlet (0.1518)	kqRWallFunction	cyclic	symmetryPlane
omega [1/s]	280	fixedValue 280	inletOutlet (280)	omegaWallFunction	cyclic	symmetryPlane
nut [m²/s]	0	calculated	calculated	nutUSpaldingWallFunction	cyclic	symmetryPlane
alphat [kg/m/s]	0	calculated	calculated	compressible::alphatWallFunction (Prt 0.85)	cyclic	symmetryPlane

Inlet turbulence: intensity `I = 5 %`, length scale `L = 0.1·d_o = 2.54 mm`, giving `k = 1.5·(I·U_in)² = 0.1518 m²/s²` and `omega = √k / (C_μ^0.25·L) = √0.1518 / (0.09^0.25 · 0.00254) ≈ 280 s⁻¹`.

Wall treatment: low-Re, wall-resolved. `nutUSpaldingWallFunction` + `omegaWallFunction` are the k-ω SST **blended** wall functions, valid continuously from the viscous sublayer through the log layer — so they remain correct even where local y^+ exceeds the ≤2 target (see §7.5). Turbulent Prandtl number `Prt = 0.85`.

4.2 Thermophysical properties (`constant/thermophysicalProperties`)

` thermoType: hePsiThermo / pureMixture / sutherland transport / janaf thermo / perfectGas equationOfState / sensibleEnthalpy molWeight = 28.96 g/mol (air) → R = 287 J/kg·K Sutherland: μ = As·T^1.5/(T + Ts), As = 1.458e-6, Ts = 110.4 K JANAF cp: Tlow 200 / Thigh 2500 / Tcommon 1500 K highCpCoeffs (3.09 0.00124 -4.2e-7 6.7e-11 -3.9e-15 -996 5.34) lowCpCoeffs (3.57 -7.2e-4 1.66e-6 -6.6e-10 5.1e-14 -1047 3.72) ` **Critical numerical note:** `Tcommon` was raised to **1500 K** — above the whole operating+transient band (740–1033 K) — deliberately. The two JANAF coefficient sets are ~1 % discontinuous in enthalpy at their join; with the default `Tcommon = 1000 K` sitting *inside* the operating range, the `T(h)` Newton inversion oscillated across that jump and failed to converge. Pushing `Tcommon` above the band makes the entire domain use one smooth coefficient set.

4.3 Turbulence model (`constant/turbulenceProperties`)

RANS, `kOmegaSST`, `turbulence on`, `printCoeffs on`.

4.4 Temperature clipping (`constant/fvOptions`)

`limitTemperature`, `selectionMode all`, **min 300 K / max 1200 K**. This is a *transient-safety* limiter that stops an early SIMPLE iterate from driving a cell outside the JANAF thermo range; at convergence the field is physical (max wall-adjacent T ≤ 1033 K) and the limiter is dormant.

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5. Numerical method and solver

5.1 Solver and schemes (`rhoSimpleFoam`; `system/fvSchemes`, `system/fvSolution`)

Steady compressible pressure-based SIMPLE.

- **ddt:** `steadyState`.
- **grad:** `Gauss linear`; `cellLimited Gauss linear 1` on U, e, h, k, omega.
- **div:** `bounded Gauss linearUpwind grad(U)` for momentum/energy (U, e, h, K, Ekp);

`bounded Gauss upwind` for k, omega, and `div(phiv,p)`.

- **laplacian:** `Gauss linear corrected`. **snGrad:** `corrected`. **wallDist:** `meshWave`.
- **Linear solvers:** `p` → GAMG / GaussSeidel, tol 1e-8, relTol 0.05; `(U|e|h|k|omega)` → PBiCGStab

/ DILU, tol 1e-8, relTol 0.1.

- **SIMPLE:** `consistent yes` (SIMPLEC), `nNonOrthogonalCorrectors 2`, residualControl 1e-4 on

p, U, e, h, k, omega.

- **Relaxation:** fields p 0.3, rho 0.05; equations U 0.5, e 0.5, h 0.5, (k|omega) 0.5.

5.2 Run configuration (`system/controlDict`)

`application rhoSimpleFoam`; `endTime 2000`; `deltaT 1`; `writeInterval 100`; `purgeWrite 2`; `writeFormat ascii`; `writePrecision 8`; `runTimeModifiable true`.

Function objects (in-run diagnostics):

- `yPlus1` — y^+ on all walls at write time.

- `wallHeatFlux1` — wall heat flux + integral heat rate `Q` over the `tube` patch (this is the primary measurement for `h`).
- `fieldMinMax1` — min/max of U, p, T every 50 steps (physicality check).
- `pIn` / `pOut` — area-averaged static pressure on `inlet` / `outlet` every step (for the streamwise Δp / friction factor).

5.3 Parallelisation (`system/decomposeParDict`)

`numberOfSubdomains 10`, `method scotch`. The ~3.29 M-cell mesh is impractical serially; the solve runs on 10 cores.

5.4 Execution pipeline (`run/Allrun`, `run/of.sh`)

All OpenFOAM commands run inside the ESI image `opencfd/openfoam-default:2406` via Docker, with the `cfd/` tree mounted at `/cfd`. **Docker quirk:** the image ENTRYPOINT `cd`s to `$HOME`, so `docker run -w` is ignored; `of.sh` instead `cd`s to the case dir *inside* the container command.

` blockMesh → surfaceFeatureExtract # sharp fin edges → tube.eMesh → snappyHexMesh -overwrite # carve tube, snap, 8 boundary layers → checkMesh # quality gate → "Mesh OK" → cp -r 0.orig 0 → decomposePar # 10 subdomains (scotch) → mpirun -np 10 rhoSimpleFoam -parallel → reconstructPar -latestTime `

5.5 Convergence (measured — solver logs)

SIMPLE **converged in 464 iterations**, all monitored residuals $< 1 \times 10^{-4}$. Final-time fields (Time 464/466) are steady; the temperature limiter is dormant (physical field).

y^+ on the `tube` patch (measured, converged): **min 6.60×10^{-4}, mean 0.244, max 4.63.** The mean satisfies the plan's $y^+ \leq 2$ wall-resolution gate; the max exceeds it locally at the fin leading-edge tips (§7.5).

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6. Post-processing, validation, and the UA(ṁ) law

6.1 Measured CFD quantities (converged, Time 466)

Frozen as documented constants in `validation/single_point_check.py` (they match the live `postprocessing.extract` output on the case; the `postProcessing/` tree itself is gitignored):

Quantity	Symbol	Value	Source
Integral wall heat rate over tube	Q	**256.30 W**	`wallHeatFlux1` integral column
Measured wetted area of tube patch	A_wetted	**9.0165×10^{-3} m²**	329 016 faces (fin-clipped)
Streamwise pressure drop	`Δp = p_in − p_out`	**148.4 Pa** (200148.4 − 200000.0)	pIn/pOut probes
Wall / inlet temperature	T_wall / T_in	1033 / 740 K	BC (input)

Inlet velocity / area	U_in / A_inlet	6.3623 m/s / 5.773×10^{-4} m^2	BC / mesh
y^+ mean / max (tube)	—	0.244 / 4.63	yPlus1
Streamwise tube rows in domain	N_rows	1	single-tube unit cell

6.2 Reduction to Nu and f (the validation, `validation/single_point_check.py`)

Property model (stated explicitly, reused everywhere for consistency): $\rho = p/(R\cdot T)$ with R = 287; μ = Sutherland; $cp = 1080$ J/kg·K (representative high-T air, JANAF band); $Pr = 0.69$ as an explicit choice so $k_{air} = \mu\cdot cp/Pr$ (rather than a derived number) keeps $Nu = h\cdot d_o/k$ consistent with the Pr fed to Briggs–Young. Properties evaluated at the **film temperature** $T_{film} = (T_{wall} + T_{bulk\_mean})/2$.

Reduction steps:

- $\rho_{in} = p/(R\cdot T_{in})$, $\dot{m} = \rho_{in}\cdot U_{in}\cdot A_{inlet}$.
- $T_{out} = T_{in} + Q/(\dot{m}\cdot cp)$; $T_{bulk\_mean} = (T_{in}+T_{out})/2$; $T_{film} = (T_{wall}+T_{bulk\_mean})/2$.
- Area-averaged wall flux $q'' = Q/A_{wetted}$.
- **Two ΔT conventions for h** (the choice is material — see below):
  - $h_{inlet} = q''/(T_{wall} - T_{in}) \rightarrow Nu_{inlet} = 40.3$
  - $h_{LMTD} = q''/LMTD$, $LMTD = (\Delta T_{in} - \Delta T_{out})/\ln(\Delta T_{in}/\Delta T_{out}) \rightarrow Nu_{LMTD} = 45.9$
- Reynolds number at the minimum-flow section: $A_{min} = 3\cdot min\_flow\_area\_per\_pitch = 2.688 \times 10^{-4}$ m^2,

$G_{max} = \dot{m}/A_{min}$, $Re = G_{max}\cdot d_o/\mu(T_{film}) = 8368$.

- Friction factor (Euler number per row): $f_{CFD} = \Delta p/(0.5\cdot\rho_{in}\cdot U_{max}^2\cdot N_{rows}) = 1.688$.

Validation against published correlations at Re = 8368:

- **Briggs & Young (1963)** heat transfer:

$Nu = 0.134\cdot Re^{0.681}\cdot Pr^{(1/3)}\cdot(s/fin_h)^{0.2}\cdot(s/fin_t)^{0.1134}$, $s = fin\_pitch - fin_t$ (inter-fin gap). → $Nu_{BY} = 54.1$.

- **Robinson & Briggs (1966)** friction (Euler number per row):

$f = 9.465\cdot Re^{(-0.316)}\cdot(S_T/d_o)^{(-0.927)}$. → $f_{RB} = 0.287$.

Quantity	CFD	Correlation	Deviation	Verdict
Nu (LMTD)	**45.9**	54.1 (Briggs–Young)	**−15.2 %**	within ±20 % → **heat transfer VALIDATED**
Nu (inlet ΔT)	40.3	54.1	−25.5 %	outside ±20 % (wrong ΔT convention)
f (friction)	**1.688**	0.287 (Robinson–Briggs)	**+489 %**	outside ±20 % → **does NOT validate**

Why LMTD, not inlet-ΔT: the air bulk-warms appreciably across the cell, so the log-mean driving ΔT is the physically correct mean for an integrated HX balance and is the basis of the Briggs–Young correlation itself. Using the simpler wall-minus-inlet ΔT understates the true driving ΔT and drops Nu to 40.3 (−25.5 %, outside the gate). **The ΔT convention is material; LMTD is the correct choice.**

Why friction misses (and why it doesn't matter): the CFD Δp spans the *whole* unit cell (inlet plenum + tube + outlet wake, ≈1.7 velocity heads), whereas Robinson–Briggs describes only the *incremental per-row* bundle loss in a fully-developed multi-row bank — different measurements. Fin clipping (§7.1) adds spurious form drag on the CFD side. Heat transfer validates cleanly because it is dominated by the well-resolved bulk boundary layer (mean y^+ 0.244), not by entrance/exit geometry. **Friction is kept informational only and is NOT wired into the v2 power balance.**

6.3 The CFD-validated UA(ṁ) law (`postprocessing/fit.py` → `results/cfd_correlation.json`)

Heat transfer validated, so the **now-CFD-validated Briggs–Young `Nu(Re)` law** is adopted as the air-side model and converted to an overall heater conductance:

`UA(ṁ) = UA_des · (Nu(ṁ)/Nu_des) · (η_o(h(ṁ))/η_o(h_des))`, with Re tracking ṁ linearly (`Re = Re_des·ṁ/ṁ_des`), **anchored to v1.5's design magnitude so v2 == v1.5 exactly at design:**

- `UA_des = 152.35 kW/K` at `ṁ_des = 54.37 kg/s` (from `cycle_model.hx_entu_v1_5.size_design`, computed live).
- Overall surface efficiency `η_o = 1 − (A_fin/A_tot)·(1 − η_fin(h))`, with `η_fin(h)` the annular fin efficiency (§2.2). Because the reference fin is thermally **inefficient** (`η_fin ≈ 0.53` at design — realistic for a low-conductivity k ≈ 25 W/m·K high-temperature alloy fin), `η_o` *falls* as h (and hence ṁ) rises: **0.62 → 0.54** across the swept range, `dln η_o/dln ṁ ≈ −0.23`.

Effective exponent (log-log fit of the anchored law over ṁ = 0.7–1.3·ṁ_des): `n_air_effective = 0.454`. The raw Nu scaling is steeper (`Nu ∝ Re^0.681`), but the fin-efficiency modulation dominates: `0.681 − 0.23 ≈ 0.454`. **This is SHALLOWER than v1.5's assumed 0.6**, not the steeper 0.66–0.68 originally anticipated.

`results/cfd_correlation.json` (the emitted interface artifact, consumed by `cfd_to_v2.py`):

Key	Value	Meaning
`C_nu`	0.13059	geometry coefficient in `Nu = C_nu·Re^m·Pr^(1/3)`
`m`	0.681	Briggs–Young Re-exponent
`C_f, p`	4.9781, −0.316	Robinson–Briggs friction (informational)
`Re_des`	8368	design/validation Reynolds number
`Pr`	0.69	Prandtl (modelling choice)
`k_air_des`	0.061133 W/m·K	air conductivity at design film T
`UA_design_kW_per_K`	152.349	v1.5 anchor
`mdot_des_kg_s`	54.370	v1.5 anchor
`n_air_effective`	0.4539	effective UA∝ṁ^n (vs v1.5's 0.6)
`n_tubes`	1281.4	**diagnostic only** (tubes at L = 2.0 m reproducing UA_des)
`validation`	{Nu 45.9/54.1/−15.2%, f 1.688/0.287/+489%}	frozen single-point outcome

`dp_of_mdot` (Robinson–Briggs Δp ∝ f(Re)·(ṁ/ṁ_des)²) is written to the JSON as **informational only** — not used in the power balance.

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7. Results

7.1 The v2 derating curve (`cfd_to_v2.py` → `results/derating_curve_v2.csv`)

The CFD-validated `UA(ṁ)` law is injected into v1.5's cycle model (replacing `ua_law`) and the derating curve is regenerated over 25–55 °C. Key rows:

Ambient	ṁ (kg/s)	TIT (°C)	Regime	net MWe	% of 25 °C	plant η
25 °C	54.370	742	B	5.000	100.0 %	0.333
35 °C	53.480	762	B	4.926	98.5 %	0.328
45 °C	52.633	782	A	4.852	97.1 %	0.323
55 °C	51.825	783	A	4.566	91.3 %	0.304

- **Mean electric derating 0.15 %/°C (25–45 °C); penalty at 55 °C = −8.7 %.**
- Air mass flow varies only ~4–5 % over the full 25→55 °C range (54.37 → 51.83 kg/s) — this is *why*

the UA-exponent change barely moves the curve.

- CSV columns: `ambient_temp_C, mdot, TIT_C, T3_C, regime, T_hp_req_C, q_cycle_MW, q_shed_MW,

net_MWe, plant_efficiency, cycle_efficiency, net_MWe_frac_vs25C. (Regime B = fixed-TIT design regime; Regime A = the hotter regime where the required heat-pipe temperature is capped and heat is shed – q_shed_MW > 0` appears at 45–55 °C.)

7.2 v2 vs v1.5 (`results/derating_curve_v1_5_vs_v2.png`)

	penalty @55 °C	net MWe @55 °C
v1.5 (assumed UA ∝ $\dot{m}^{0.6}$)	−8.8 %	4.57
v2 (CFD Briggs–Young, n ≈ 0.454)	**−8.7 %**	4.57
Δ (v2 − v1.5) @55 °C	**+0.08 pt**	≈ 0

v2 CONFIRMS v1.5. The exponent change 0.6 → 0.454 shifts the 55 °C penalty by less than one tenth of a percentage point. The finding worth reporting *is* this robustness: the derating does not hinge on the fine detail of the heater's heat-transfer law; it is set by the open-air Brayton's compressor-inlet ("hot-day") thermodynamics, which v1/v1.5 already captured. An independent physics simulation confirming that assumption raises confidence in the whole Phase-1 result.

7.3 Output artifacts

File	Contents
`phase1_derating/results/derating_curve_v2.csv`	the v2 curve (25–55 °C, 1 °C steps)

phase1_derating/results/derating_curve_v1_5_vs_v2.png	v1.5 vs v2 comparison plot
phase1_derating/cfd/results/cfd_correlation.json	the CFD-validated UA(ṁ) law + validation numbers

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8. Honest limitations

- **Fin clipping.** `r_f = 24.7 mm > S_L/2 = 22.2 mm`, so the 12 mm fins overrun the single-tube streamwise box and are clipped at the inlet/outlet planes (~42 % of the inlet plane is fin metal at fin z-levels). Inherent to the reference geometry (fin height ≈ tube radius) combined with a single-tube box — one cell cannot represent the interleaving of staggered neighbours' fins. It biases the **friction** measurement (spurious clipped-fin form drag); heat transfer still validates (−15.2 %).
- **Friction not validated, not used.** The Robinson–Briggs Δp / parasitic-power estimate is carried in `cfd_correlation.json` for reference only.
- **Mesh independence not formally studied.** A single validated point stands in for a grid-convergence study, supported by *adequacy evidence* (−15.2 % Nu agreement, mean y^+ 0.244, clean checkMesh) rather than proof. A formal 3-level grid-convergence study is future work.
- **ω bounding at fin tips.** A localized near-singular-corner artifact in the k-ω solve at the fin leading edges; does not perturb the converged U/p/T fields (integrated h/f are bulk-dominated).
- **y^+ gate exceedance at fin leading-edge tips.** Plan gate is $y^+ \leq 2$ on tube+fin walls. Mean y^+ (0.244) satisfies the wall-resolved intent; max (4.63) exceeds it locally at the fin tips (the same corners behind #4). The k-ω SST blended wall functions (nutUSpalding/omega) remain valid at these y^+, and integrated h is bulk-dominated, so this does not compromise the validation.
- **Option 1, not a full sweep.** Single-point validation + correlation-driven curve, not a 6-point CFD Reynolds sweep. Chosen once the point validated heat transfer within the plan's own ±20 % gate. A fully CFD-derived `Nu(Re)` correlation and a friction-domain fix (dedicated Δp planes bracketing the bundle, or a multi-row domain) are documented future extensions.

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9. How to reproduce

`bash

--- CFD (needs Docker + the ESI OpenFOAM v2406 image) ---

cd phase1_derating/cfd python3 -c "from geometry.finned_tube import REFERENCE; \ from geometry.make_stl import write_unitcell_stl; \ print(write_unitcell_stl(REFERENCE, 'heater_unitcell/constant/triSurface/tube.stl'))" run/Allrun # blockMesh → snappy → checkMesh → parallel rhoSimpleFoam → reconstruct

(~25 min on 10 cores; mesh/logs/processor dirs are gitignored)

--- Post-processing / validation / injection (pure Python, no Docker) ---

python3 -m validation.single_point_check # CFD vs Briggs-Young/Robinson-Briggs at Re=8368 python3 -m postprocessing.fit # writes results/cfd_correlation.json python3 cfd_to_v2.py # injects UA law into v1.5 → derating_curve_v2.csv + png python3 -m pytest -q # 21-test suite (geometry/correlations/extract/fit/...) `

What is version-controlled vs regenerated: the OpenFOAM case *source* is committed — `system/` dicts, `constant/{thermophysicalProperties,turbulenceProperties,fvOptions}`, `0.orig/` fields, and `constant/triSurface/tube.stl`. Run artifacts (`0/`, `processorN/`, `constant/polyMesh/`, `postProcessing/`, logs) are **gitignored** and regenerated by `run/Allrun`. The frozen measured CFD scalars live as documented constants in `validation/single_point_check.py` and `postprocessing/fit.py`, so the Python validation/injection reproduces without re-running CFD.

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10. Future work (in priority order)

- **Intake/exhaust recirculation study** — the highest-value next step. In a real desert skid, hot exhaust can be pulled back into the intake, raising the effective inlet temperature above ambient and making derating *worse*. No correlation covers this (it depends on skid layout + local wind), so it genuinely needs CFD. The validated air-side setup here is the foundation. Feeds Phase-2 siting (layout-dependent penalty). Needs an assumed skid external geometry.
- **Full multi-condition CFD Reynolds sweep** — derive `Nu(Re)` entirely from simulation instead of validating the correlation.
- **Fin-geometry sensitivity** — the reference fins are thermally inefficient (η_fin ≈ 0.53); better fins would change the scaling.
- **Formal 3-level mesh-independence study** — publication-grade rigour on the single validated point.
- **Friction-domain fix** — dedicated Δp planes bracketing the bundle, or a multi-row domain, so friction can be validated and the parasitic-power estimate wired into the power balance.

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11. Provenance and cross-references

Topic	Location
Reproduction detail + directory layout	`phase1_derating/cfd/README.md`
Decision record & full rationale (Option 1)	`DECISIONS_LOG.md` entry **D12**

Reference-design facts + v2 numbers (spec form)	`phase1_derating/spec/A1_reference_spec_sheet.md` §4d
Geometry code	`phase1_derating/cfd/geometry/{finned_tube.py, make_stl.py}`
Correlations	`phase1_derating/cfd/correlations/finned_tube_corr.py`
Validation	`phase1_derating/cfd/validation/single_point_check.py`
Post-processing / UA law	`phase1_derating/cfd/postprocessing/{extract.py, fit.py}`
v2 injection	`phase1_derating/cfd/cfd_to_v2.py`
The curve (interface artifact)	`phase1_derating/results/derating_curve_v2.csv`, `derating_curve_v1_5_vs_v2.png`

Correlation references:

- Briggs, D.E. & Young, E.H. (1963), *Convection heat transfer and pressure drop of air flowing across triangular pitch banks of finned tubes*, Chem. Eng. Prog. Symp. Ser.
- Robinson, K.K. & Briggs, D.E. (1966), *Pressure drop of air flowing across triangular pitch banks of finned tubes*, Chem. Eng. Prog. Symp. Ser.

Software: OpenFOAM (ESI) v2406, image `opencfd/openfoam-default:2406`. Python post-processing uses numpy / pandas / matplotlib / numpy-stl.